

Energy-Efficient 3-D UAV Ground Node Accessing Using the Minimum Number of UAVs

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Abstract—Cooperative multiple unmanned aerial vehicles (UAVs) have been widely exploited in various applications, including data collection, forest monitoring, edge computing, and so on. Due to limited onboard storage and expensive hardware costs, reducing both the energy consumption and the number of UAVs is critical for these multi-UAV applications. However, existing studies primarily revolved around energy minimization in two-dimensional (2-D) scenarios, given a sufficient but fixed number of UAVs, and most of them considered specific application scenarios, resulting in poor generality. In contrast, this paper defines a generalized application scenario, in which multiple ground nodes (GNs) are accessed by multiple UAVs in three-dimensional (3-D) scenarios, and aims to minimize the energy consumption by employing the necessary (or equivalently minimum) number of UAVs and formulating a mix-integer nonconvex problem. To this end, this paper decomposes the problem into two subproblems: energy consumption minimization for a single UAV consecutively accessing any two GNs and energy-efficient multi-UAV GN-accessing path planning employing the minimum number of UAVs. The first subproblem is solved by applying the successive convex approximation (SCA) technique and the path discretization method, while the second subproblem is addressed by designing a three-stage approximation framework based on modified particle swarm optimization (MPSO) and greedy path assignment (GPA). Comprehensive simulations demonstrate the superior performance of the proposed method in terms of optimality and efficiency compared to several other counterparts.

Index Terms—Multiple UAVs, energy consumption, number of UAVs, GN-accessing, MPSO, GPA.

I. INTRODUCTION

IN THE past few years, unmanned aerial vehicles (UAVs), especially multi-rotor UAVs, have experienced an unprecedented level of growth due to their great mobility. There emerge

many of their applications, such as search and rescue [1], [2], data collection [3], [4], cargo delivery [5], public safety [6], [7], and edge computing [8]. Multi-UAV cooperation plays an important role in these applications, which has triggered numerous studies dedicated to minimizing the energy consumption and the number of UAVs due to limited onboard storage and hardware costs. However, these studies often employ complex constraints for specific scenarios to minimize the UAV energy consumption, thereby restricting the wide applications of relevant optimization schemes and narrowing the scope for UAV energy minimization.

Existing UAV studies have highlighted the importance of minimizing the energy consumption and the number of UAVs [9], [10]. First, due to the limitation of onboard storage capacity, energy consumption minimization is beneficial to improve the range of UAVs. [11], [12], [13] minimized the energy consumption by a trajectory optimization method, but this method is only applicable to single-rotor UAVs in two-dimensional (2-D) scenarios. In [14], [15], [16], the authors solved the energy consumption minimization problem for UAVs in three-dimensional (3-D) scenarios but did not consider the case of multiple UAVs. Therefore, great efforts have to be devoted to minimizing the 3-D flight energy consumption of multiple UAVs, especially popular multi-rotor UAVs. Second, due to the expensive purchase and maintenance costs of UAVs, minimizing the number of UAVs can avoid additional overhead. However, in most existing studies [17], [18], [19], the number of UAVs was considered to be fixed (usually sufficient) rather than minimum. Although [8] minimized the number of UAVs, the UAV positions were fixed, restricting its broader applicability in multi-UAV scenarios. Thus, a more adaptable optimization method for UAV number minimization is warranted.

In summary, there are two shortcomings in the above studies. First, these optimization schemes regarding the energy consumption or the number of UAVs lack generality. Second, to the best of our knowledge, no related work has addressed the more practical problem of minimizing the total energy consumption while employing the minimum number of UAVs for task fulfillments.

Consequently, to develop more applicable optimization schemes and to explore in depth the energy-efficient performance of UAVs, this paper first generalizes current popular multi-UAV application scenarios and defines a generic multi-UAV application scenario, termed ground node (GN) accessing scenario, as shown in Fig. 1. Concretely, in this scenario, multiple UAVs are scheduled to depart from a depot to access a set of GNs and return to the same depot. GNs are accessed only

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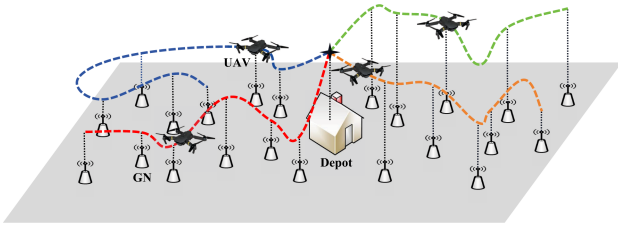


Fig. 1. An overview of the GN-accessing scenario.

when UAVs are directly above them, diminishing the impact of factors such as communication range on UAV trajectories, and in this way effectively harnesses trajectory planning to fully explore the potential space for UAV energy consumption minimization.

On this basis, this paper formulates a problem of minimizing the energy consumption for GN-accessing tasks while employing the minimum number of UAVs. Given the complexity of this mix-integer nonconvex problem, it is divided into two subproblems for ease of resolution. To be specific, the first subproblem aims to minimize the 3-D energy consumption required for a single UAV to access any two GNs consecutively, which is formulated based on the 3-D energy consumption model for multi-rotor UAVs in our previous work [20], the successive convex approximation (SCA) technique [12] and the path discretization method [11] are applied to solve this subproblem. The second subproblem aims to plan energy-efficient multi-UAV GN-accessing paths employing the minimum number of UAVs, and an approximation algorithm integrating modified particle swarm optimization (MPSO) and greedy path assignment (GPA), namely MPSO-GPA, is proposed to solve this subproblem.

For the purpose of performance evaluation, extensive simulations were performed. It is shown that the optimized trajectory demonstrates up to a 16% reduction in UAV energy consumption compared to a linear trajectory, confirming the effectiveness of the SCA-based trajectory optimization method. In addition, the superiority of the proposed approximation algorithm for minimizing the energy consumption and the number of UAVs is demonstrated by comparing it with benchmarks. The feasibility of the overall algorithms is further emphasized by analyzing the optimization results of energy-efficient multi-UAV GN-accessing.

To sum up, the main contributions of this work are summarized as follows.

- In the context of the designed multi-UAV GN-accessing scenario, an optimization problem aiming to minimize the energy consumption while employing the minimum number of UAVs is formulated. This problem is proved to be mix-integer nonconvex, and is decomposed into two subproblems to solve.
- The SCA-based algorithm is applied to solve the first subproblem, i.e., minimizing the 3-D flight energy consumption required for a multi-rotor UAV to access any two GNs consecutively. Without loss of generality, this

algorithm is broadly applicable to other energy-efficient multi-rotor UAV applications.

- The approximation framework MPSO-GPA is developed to solve the second subproblem, i.e., planning energy-efficient multi-UAV GN-accessing paths by employing the minimum number of UAVs. This algorithm is also generic enough to be applied to other economical multi-UAV path planning.
- Extensive simulations are conducted to evaluate the utility of the proposal, and the results demonstrate that the proposed methods outperform benchmarks.

II. RELATED WORK

In this section, literature on energy consumption minimization in common UAV-assisted applications including data collection, wireless communication, and edge computing is briefly introduced.

In data collection applications, [21] applied classic linear programming and heuristic algorithms to design energy-efficient path planning for data collection. In [22], the task area was segmented into grids, each with a central wireless charging device for UAVs; reinforcement learning was utilized to optimize the selection of charging points, flying altitude, and connected sensor nodes, thereby enhancing UAV energy efficiency. Furthermore, as mentioned in [23], a fixed-wing UAV was deployed for secure data collection, minimizing the combined energy consumption of the UAV and sensor devices (SDs) by jointly optimizing the UAV's 3D trajectory and the SDs' transmission schedule. [24] demonstrated a strategy involving multiple UAVs with pre-planned trajectories to collect sensory data from stationary devices, effectively reducing energy consumption.

In wireless communication applications, to overcome the limitations of onboard batteries for UAV-aided communication applications, [25] formulated an optimization problem aiming to minimize the UAV's energy procurement costs. The authors in [26] treated the UAV placement issue as a constrained optimization problem, aiming to optimize for maximum equitable coverage and minimum energy consumption while adhering to backhaul requirements at specific time nodes. In [13], the authors focused on establishing the energy model for single-rotor UAVs, and under the premise of limited UAV energy, they jointly optimized sensor communication scheduling, transmit power allocation, and UAV trajectory to minimize the maximum energy consumption of the UAV and sensors. Moreover, [27] investigated the problem of scheduling the movement of multiple UAVs to fairly provide energy-efficient communication services to mobile ground users for a given period, by using the deep reinforcement learning (DRL) methods.

In edge computing applications, [28] studied the energy consumption in terminals on the ground and UAVs and came up with the idea of UAV trajectory adjustment in solving the trade-off brought by flight propulsion. In [29], the authors proposed an attention-based multi-agent proximal policy optimization (MAPPO) algorithm to pursue the optimal computation offloading policy of UAVs to minimize the weighted energy consumption while considering service fairness. To minimize the

TABLE I
CONTRIBUTION COMPARISONS OF PREVIOUS WORKS AND THIS WORK

Elements	this work	[17]	[26]	[27]	[28]	[29]	[30]	[31]	[32]	[33]	[34]	[35]	[36]
Multi-UAV	✓	×	×	×	×	✓	×	✓	×	×	✓	✓	✓
3-D mobility	✓	×	×	×	✓	×	✓	×	✓	✓	✓	✓	×
Multi-rotor UAV	✓	×	✓	✓	×	—	×	✓	✓	×	×	✓	✓
Accurate energy model	✓	✓	✓	×	✓	✓	✓	✓	×	✓	✓	×	✓
Minimum UAV number	✓	×	×	×	×	×	×	×	×	×	×	×	×

UAV energy consumption, [30] introduced a cellular-connected UAV mobile edge computing (MEC) system, optimizing bit allocation, power allocation, and the trajectories of multiple UAVs. In [31], to reduce the energy consumption, the authors optimized the trajectory planning of UAVs and computation resource allocation by converting these two problems into convex ones, and solved them with an efficient iterative algorithm.

Nevertheless, the above research suffers from the following two critical limitations. First, current UAV energy consumption minimization schemes are typically tailored to specific scenarios and struggle to tackle problems that simultaneously incorporate common elements like multi-rotor UAVs, multi-UAV cooperation, 3-D mobility, and accurate energy consumption models, thereby restricting their broader applicability. Second, the number of UAVs in multi-UAV-assisted applications is often predetermined, not minimized, resulting in extra hardware costs. Table I delineates the differences between this paper and the prior studies. Hence, it is imperative to develop a generalized multi-UAV application scenario in which a thorough UAV energy consumption minimization scheme is designed while taking into account UAV number minimization.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section first introduces the system model and the UAV energy consumption model in multi-UAV GN-accessing scenarios, and then formulates an optimization problem aimed at minimizing the energy consumption for GN-accessing tasks while employing the minimum number of UAVs; recognizing the high complexity of direct solutions, this problem is efficiently decomposed into two subproblems, finally, a detailed examination of approximation executions' feasibility in optimization problems is given. To clarify, Fig. 2 illustrates the optimization problems and their main decision variables in this paper.

A. System Model

Considering a GN-accessing scenario in which $M \in \mathbb{Z}^+$ homogeneous multi-rotor UAVs are assigned to access $N \in \mathbb{Z}^+$ GNs distributed over a task space, where a GN can be a wireless sensor, a facility, or a mobile device. Assume that M UAVs are deployed, where the value of M is unknown and will be determined later. The UAVs and GNs are denoted by the sets $\mathcal{M} = \{1, \dots, M\}$ and $\mathcal{N} = \{1, \dots, N\}$, respectively. The horizontal location and the access altitude of the n th GN with $n \in \mathcal{N}$, denoted as $\mathbf{w}_n \in \mathbb{R}^{2 \times 1}$ and H_n , respectively, are pre-known before task assignment. All UAVs are assumed to depart from the same depot at the same time and return to that depot after the task is fulfilled, namely that all GNs are accessed. Let T denote the total time for the UAVs to fulfill the task. Moreover, for

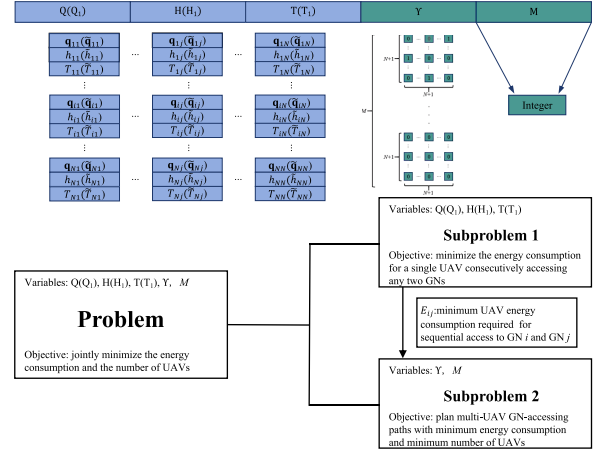


Fig. 2. Problem decomposition and main decision variables.

harmonization, the horizontal location of the depot is denoted as $\mathbf{w}_0 \in \mathbb{R}^{2 \times 1}$, and the flight altitude of the UAVs departing from or to the depot is defined as H_0 . The overall scenario of GN-accessing tasks is shown in Fig. 1.

B. The Generic Energy Consumption Model

In this subsection, a generic power consumption model for multi-rotor UAVs is given and further applied to calculate 3-D UAV energy consumption.

Simplifying UAV power consumption models, while ensuring their reasonableness, significantly enhances the efficiency of energy consumption minimization. Since communication-related energy consumption is significantly less than propulsion energy consumption [32], and acceleration/deceleration energy consumption forms a minor part of overall energy consumption [11], the two are ignored. To this end, an efficient 3-D propulsion power consumption model for UAVs in steady status [20], which is mainly determined by the horizontal velocity $\mathbf{V}^{\parallel}$ and the vertical velocity $\mathbf{V}^{\perp}$, is presented as (1) shown at the bottom of the next page, where sgn is a sign function and other parameters are described in Table II.

On that basis, the corresponding 3-D propulsion energy consumption of multi-rotor UAVs, denoted as E_1 , can be heuristically modeled as

$$E_1(\{\mathbf{q}(t)\}, \{h(t)\}, T^t) = \int_0^{T^t} P_{total}(\mathbf{V}^{\parallel}(t), \mathbf{V}^{\perp}(t)) dt, \quad (2)$$

where $\mathbf{q}(t) \in \mathbb{R}^{2 \times 1}$ and $h(t)$ define the horizontal and vertical coordinates corresponding to the UAV trajectory at time instant

TABLE II
MAIN NOTATIONS OF THE ENERGY CONSUMPTION MODEL

Notation	Physical meaning	Simulation value
n_r	Number of rotors	4
δ	Profile drag coefficient	0.011
ρ	Air density in kg/m ³	1.168
s	Rotor solidity	0.045
A	Rotor disc area in m ²	0.214
W	UAV weight in Newton	20
v_0	Mean rotor induced velocity in hover	6.325
C_t	Thrust coefficient	0.001195
$S_{FP\parallel}$	Fuselage equivalent flat plate area in horizontal status in m ²	0.009
$S_{FP\perp}$	Fuselage equivalent flat plate area in vertical status in m ²	0.19
P_{bl}	Blade profile power for UAVs in hovering status	133.9869
P_{in}	Induced power for UAVs in hovering status	70.2132

t , respectively, T^t is the UAV flight time, moreover, $\mathbf{V}^\parallel(t)$ and $\mathbf{V}^\perp(t)$ satisfy $\mathbf{V}^\parallel(t) \triangleq \dot{\mathbf{q}}(t)$ and $\mathbf{V}^\perp(t) \triangleq \dot{\mathbf{h}}(t)$, respectively.

C. Problem Formulation

Regarding the optimization problem of minimizing the energy consumption for GN-accessing tasks while employing the minimum number of UAVs, this subsection first gives the decision variables and constraints related to the optimization problem, and then formulates the optimization problem in closed form.

1) *Decision Variables and Constraints*: First, to describe the association status and sequence of the m th UAV among the GNs, a binary decision $x_{ij}[m]$, $i, j \in \mathcal{N} \cup \{0\}$ is introduced, which equals 1 if the m th UAV accesses GN (or depot) j after GN (or depot) i , and equals 0 otherwise. In this case, the flow constraints can be expressed as

$$\sum_{i \in \mathcal{N}} x_{i0}[m] = 1, \quad \forall m \in \mathcal{M}, \quad (3)$$

$$\sum_{j \in \mathcal{N}} x_{0j}[m] = 1, \quad \forall m \in \mathcal{M}, \quad (4)$$

$$\sum_{m \in \mathcal{M}} \sum_{i \in \mathcal{N}} x_{ij}[m] = 1, \quad \forall j \in \mathcal{N}, \quad (5)$$

$$\sum_{i \in \mathcal{N}} x_{iv}[m] = \sum_{j \in \mathcal{N}} x_{vj}[m], \quad \forall v \in \mathcal{N} \cup \{0\}, \quad \forall m \in \mathcal{M}, \quad (6)$$

$$x_{ij}[m] \in \{0, 1\}, \quad \forall i, j \in \mathcal{N} \cup \{0\}, \quad \forall m \in \mathcal{M}, \quad (7)$$

where constraints (3) and (4) ensure that all UAVs depart from the depot as tasks start and return to the same depot after tasks are fulfilled. Constraint (5) states that each GN is accessed once by one UAV. Constraint (6) guarantees that each UAV leaves a GN if and only if it arrives at that GN. Constraint (7) specifies the possible values $x_{ij}[m]$ can take.

Then, for the purpose of representing the set of GNs accessed by the m th UAV, $O_m \in \mathbf{O}$ is introduced to denote the corresponding tour¹ of the m th UAV, where $\mathbf{O}$ is the full set of tours and its value, denoted as $|\mathbf{O}|$, equals M . The set of GNs covered by O_m , denoted as S_{O_m} , satisfies the following condition

$$\sum_{i \in S_{O_m}} \sum_{j \in S_{O_m}} x_{ij}[m] = |S_{O_m}| - 1, \quad (8)$$

$$S_{O_m} \subseteq \mathcal{N}, \quad |S_{O_m}| > 1.$$

Moreover, the energy spent by the m th UAV on the tour O_m , namely E_{O_m} , can be expressed as

$$E_{O_m} = \sum_{i \in S_{O_m}} (x_{i0}[m]E_{i0} + x_{i0}[m]E_{i0}) + \sum_{i \in S_{O_m}} \sum_{j \in S_{O_m}} x_{ij}[m]E_{ij}, \quad (9)$$

where E_{ij} represents the energy consumption for a UAV consecutively accessing GN i and GN j , and is calculated as

$$E_{ij} = E_1 \left(\{\tilde{\mathbf{q}}_{ij}(t_{ij})\}, \{\tilde{\mathbf{h}}_{ij}(t_{ij})\}, \tilde{T}_{ij}^t \right) = \int_0^{\tilde{T}_{ij}^t} P_{total} \left(\tilde{\mathbf{V}}_{ij}^\parallel(t_{ij}), \tilde{\mathbf{V}}_{ij}^\perp(t_{ij}) \right) dt_{ij}, \quad (10)$$

where $\tilde{\mathbf{q}}_{ij}(t_{ij})$, $\tilde{\mathbf{h}}_{ij}(t_{ij})$, $\tilde{\mathbf{V}}_{ij}^\parallel(t_{ij})$ and $\tilde{\mathbf{V}}_{ij}^\perp(t_{ij})$ define the horizontal coordinates, vertical coordinates, horizontal velocity, and vertical velocity of a UAV at instant time t_{ij} under the flight trajectory between consecutively accessed GNs i and j ,

¹A tour is a feasible and complete path for a UAV to cover all GNs assigned to it.

$$P_{total}(\mathbf{V}^\parallel, \mathbf{V}^\perp) = P_{bl} + \frac{3}{8} \sqrt{n_r} \delta \sqrt{\frac{W \rho A}{C_t}} s \|\mathbf{V}^\parallel\|^2 + P_{in} \left(\sqrt{1 + \frac{\|\mathbf{V}^\parallel\|^4}{4v_0^4}} - \frac{\|\mathbf{V}^\parallel\|^2}{2v_0^2} \right)^{1/2} + \frac{n_r}{2} S_{FP\parallel} \rho \|\mathbf{V}^\parallel\|^3 + \frac{1}{2} W \|\mathbf{V}^\perp\| + sgn(\mathbf{V}^\perp) \frac{n_r}{4} S_{FP\perp} \rho \|\mathbf{V}^\perp\|^3 + \left(\frac{W}{2} + sgn(\mathbf{V}^\perp) \frac{n_r}{4} S_{FP\perp} \rho \|\mathbf{V}^\perp\|^2 \right) \times \sqrt{\|\mathbf{V}^\perp\|^2 + sgn(\mathbf{V}^\perp) \frac{S_{FP\perp}}{A} \|\mathbf{V}^\perp\|^2 + \frac{2W}{n_r \rho A}} + (sgn(\|\mathbf{V}^\perp\|) - 1) \frac{W}{2} \sqrt{\frac{2W}{n_r \rho A}}, \quad (1)$$

respectively; in addition, $\tilde{T}_{ij}^t$ is the UAV flight time spent in this trajectory.

Furthermore, given the maximum energy limit $E_{\max}$ for a UAV, E_{O_m} of the m th UAV must not exceed this threshold. Thus, the following constraint should be satisfied

$$E_{O_m} \leq E_{\max}, \quad \forall O_m \in \mathbf{O}. \quad (11)$$

2) *Mathematical Formulation:* Prior to formulating the optimization problem, the structural relationship between the two key optimization metrics in it is examined. While this optimization problem emphasizes two *minimums*, i.e., the minimum total energy consumption and the minimum number of UAVs, this problem treats only the former as the optimization objective due to the UAV number being a prerequisite for calculating total energy consumption, with the latter serving as a decision variable, termed the total energy consumption minimization problem employing the minimum number of UAVs.

Consequently, based on (9), the total energy consumption minimization problem involving five optimization variables, i.e., the UAV path $\Upsilon = \{x_{ij}[m], \forall m \in \mathcal{M}, \forall i, j \in \mathcal{N}\}$, the UAV horizontal trajectory $\mathbf{Q} = \{\tilde{\mathbf{q}}_{ij}(t_{ij}), \forall i, j \in \mathcal{N}, \forall t_{ij} \in [0, \tilde{T}_{ij}^t]\}$, the UAV vertical trajectory $\mathbf{H} = \{\tilde{h}_{ij}(t_{ij}), \forall i, j \in \mathcal{N}, \forall t_{ij} \in [0, \tilde{T}_{ij}^t]\}$, the UAV flight time $\mathbf{T} = \{\tilde{T}_{ij}^t, \forall i, j \in \mathcal{N}\}$ and the UAV number M (as aforementioned, M discussed below is restricted to its lower bound, representing the minimum UAV number required for task fulfillments), can be formulated as

$$\begin{aligned} \text{(P1): } & \min_{\Upsilon, \mathbf{Q}, \mathbf{H}, \mathbf{T}, M} \sum_{O_m \in \mathbf{O}} E_{O_m} \\ \text{s.t. } & \sum_{j \in \mathcal{N}} \sum_{m \in \mathcal{M}} x_{0j}[m] = M, \\ & \text{(3)–(7), (11).} \end{aligned} \quad (12)$$

Thereby, to deal with Problem (P1), these five variables need to be determined. Fortunately, following the similar analysis in [17], given $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$, Problem (P1) can be approximated as an easily solvable multiple traveling salesmen problem (MTSP), and the proof can be found in Proposition 1. Obviously, $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$ should be prioritized, thereafter, the remaining parameters Υ and M can be tackled by solving the approximated MTSP. As such, solving Problem (P1) equates to addressing two subproblems with respect to $\{\mathbf{Q}, \mathbf{H}, \mathbf{T}\}$ and $\{\Upsilon, M\}$, respectively.

First, it can be easily known from (10) that $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$ can be obtained by minimizing E_{ij} , hence the first subproblem can be expressed as

$$\text{(P2): } \min_{\mathbf{Q}, \mathbf{H}, \mathbf{T}} E_{ij}$$

$$\begin{aligned} \text{s.t. } & \|\dot{\tilde{\mathbf{q}}}_{ij}(t_{ij})\| \leq V_{\max}^{\parallel}, \quad |\dot{\tilde{h}}_{ij}(t_{ij})| \leq V_{\max}^{\perp}, \\ & \forall t_{ij} \in [0, \tilde{T}_{ij}^t], \end{aligned} \quad (13)$$

$$\tilde{\mathbf{q}}_{ij}(0) = \mathbf{w}_i, \quad \tilde{\mathbf{q}}_{ij}(\tilde{T}_{ij}^t) = \mathbf{w}_j, \quad (14)$$

$$\tilde{h}_{ij}(0) = H_i, \quad \tilde{h}_{ij}(\tilde{T}_{ij}^t) = H_j, \quad (15)$$

where $V_{\max}^{\parallel}$ and $V_{\max}^{\perp}$ are the maximum UAV horizontal speed and maximum vertical speed, respectively.

Apparently, the first subproblem is still difficult to solve directly since it involves an infinite number of variables with respect to t_{ij} and contains a complex energy consumption cost function. In Section IV, a solution based on SCA technique and path discretization is proposed to deal with Problem (P2).

To this end, the determination of $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$ enables Problem (P1) to be further transformed into the second subproblem, which is described as

$$\begin{aligned} \text{(P3): } & \min_{\Upsilon, M} \sum_{O_m \in \mathbf{O}} E_{O_m} \\ \text{s.t. } & \text{(3)–(7), (11)–(12).} \end{aligned}$$

Second, M needs to be prioritized as a way to obtain Υ and the minimum total UAV energy consumption. As previously stated, given M , Problem (P3) transforms into a variant MTSP with determined energy weights aiming to minimize the total UAV energy consumption, yielding the path planning results as Υ . In Section V, a framework is proposed to calculate the minimum number of UAVs and further applied to solve Problem (P3).

D. Analysis of Approximation Executions in Optimization Problems

To explain the approximation executions carried out in formulating the above optimization problems, the following theoretical analysis is provided.

Proposition 1: Given $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$, Problem (P1) is closely analogous to a MTSP.

Proof: Dissecting the relationship between variables and optimization problems is crucial to simplifying optimization problems. To be specific, Problem (P1) contains five variables, i.e., $\mathbf{Q}$, $\mathbf{H}$, $\mathbf{T}$, Υ and M , where $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$ are UAV trajectory variables that directly affect UAV propulsion energy consumption, while Υ and M can determine flight paths of multiple UAVs based on the determination of these three. Therefore, given $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$, Problem (P1) is essentially approximated to an energy-efficient multi-UAV path planning problem with established energy weights (i.e., E_{ij} , $i, j \in \mathcal{N}$), which is closely analogous to a MTSP. $\square$

IV. PATH DISCRETIZATION AND SCA BASED SOLUTION FOR SOLVING PROBLEM (P2)

In this section, path discretization approaches are first applied to transform Problem (P2) into a more solvable form, and then a SCA based algorithm is proposed to obtain a suboptimal solution of Problem (P2), finally, an analysis of proposed optimization problem and algorithm is presented.

A. Path Discretization to Problem (P2)

Problem (P2) is reformulated as a more tractable discretized form by path discretization. To be specific, the path of the UAV accessing GN i and GN j sequentially is discretized into $U + 1$ line segments [11], whose horizontal and vertical projections are represented by $U + 2$ horizontal waypoints $\{\tilde{\mathbf{q}}_{ij}^u\}_{u=0}^{U+1}$ and

$U + 2$ vertical waypoints $\{\tilde{h}_{ij}^u\}_{u=0}^{U+1}$, respectively, where $\tilde{\mathbf{q}}_{ij}^0 = \mathbf{w}_i$, $\tilde{\mathbf{q}}_{ij}^{U+1} = \mathbf{w}_j$, $\tilde{h}_{ij}^0 = H_i$ and $\tilde{h}_{ij}^{U+1} = H_j$. For ease of presentation, define $\mathcal{U} = \{0, \dots, U + 1\}$. Note that $\sum_{u \in \mathcal{U}} \tilde{\mathbf{q}}_{ij}^u$ and $\sum_{u \in \mathcal{U}} \tilde{h}_{ij}^u$ are equal to $\int_0^{\tilde{T}_{ij}} \tilde{\mathbf{q}}_{ij}(t_{ij}) dt_{ij}$ and $\int_0^{\tilde{T}_{ij}} \tilde{h}_{ij}(t_{ij}) dt_{ij}$, respectively. Besides, the duration $\{\tilde{T}_{ij}^u\}_{u=0}^U$ is expressed as the time spent by the UAV on the u th line segment, the sum of which is equal to $\tilde{T}_{ij}^t$. The following constraints are imposed:

$$\|\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u\| \leq \Delta_{\max}^{\parallel}, \quad |\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u| \leq \Delta_{\max}^{\perp}, \quad \forall u \in \mathcal{U}, \quad (16)$$

where $\Delta_{\max}^{\parallel}$ and $\Delta_{\max}^{\perp}$ are appropriately chosen values to ensure that horizontal and vertical velocities of UAVs can be assumed to be constant in each line segment, and U is usually chosen to be large enough to ensure the completion of the flight. Furthermore, the UAV horizontal velocity and the UAV vertical velocity along the u th line segment are thus given by $\tilde{\mathbf{v}}_{ij}^u = \frac{\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u}{\tilde{T}_{ij}^u}$ and $\tilde{v}_{ij}^{\perp u} = \frac{\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u}{\tilde{T}_{ij}^u}$, respectively.

As a result, the optimization variables $\mathbf{Q}$, $\mathbf{H}$ and $\mathbf{T}$ are replaced by $\mathbf{Q}_1 = \{\tilde{\mathbf{q}}_{ij}^u, \forall u \in \mathcal{U}\}$, $\mathbf{H}_1 = \{\tilde{h}_{ij}^u, \forall u \in \mathcal{U}\}$, and $\mathbf{T}_1 = \{\tilde{T}_{ij}^u, \forall u \in \mathcal{U}\}$, respectively, the energy minimization Problem (P2) can be expressed in the discrete form as

$$(P2.1): \min_{\mathbf{Q}_1, \mathbf{H}_1, \mathbf{T}_1} E_1(\mathbf{Q}_1, \mathbf{H}_1, \mathbf{T}_1)$$

$$\text{s.t.} \quad \|\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u\| \leq \min \left\{ \Delta_{\max}^{\parallel}, \tilde{T}_{ij}^u V_{\max}^{\parallel} \right\}, \quad \forall u \in \mathcal{U}, \quad (17)$$

$$|\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u| \leq \min \left\{ \Delta_{\max}^{\perp}, \tilde{T}_{ij}^u V_{\max}^{\perp} \right\}, \quad \forall u \in \mathcal{U}, \quad (18)$$

$$\tilde{\mathbf{q}}_{ij}^0 = \mathbf{w}_i, \quad \tilde{\mathbf{q}}_{ij}^{U+1} = \mathbf{w}_j, \quad (19)$$

$$\tilde{h}_{ij}^0 = H_i, \quad \tilde{h}_{ij}^{U+1} = H_j, \quad (20)$$

where

$$\begin{aligned} E_1(\mathbf{Q}_1, \mathbf{H}_1, \mathbf{T}_1) = & \sum_{u=0}^U \left[P_{bl} \tilde{T}_{ij}^u + P_1 \frac{\|\Delta_{ij}^u\|^2}{\tilde{T}_{ij}^u} \right. \\ & + P_{in} \sqrt{\sqrt{\tilde{T}_{ij}^{u^4} + \frac{\|\Delta_{ij}^u\|^4}{4v_0^4}} - \frac{\|\Delta_{ij}^u\|^2}{2v_0^2}} \\ & + P_2 \frac{\|\Delta_{ij}^u\|^3}{\tilde{T}_{ij}^{u^2}} + P_3 |\Delta_{ij}^{hu}| \\ & + P_4 \frac{\Delta_{ij}^{hu^3}}{\tilde{T}_{ij}^{u^2}} + \left(P_3 + P_4 \frac{|\Delta_{ij}^{hu}| |\Delta_{ij}^{hu}|}{\tilde{T}_{ij}^{u^2}} \right) \\ & \times \sqrt{\Delta_{ij}^{hu^2} + P_5 |\Delta_{ij}^{hu}| |\Delta_{ij}^{hu}| + P_6 \tilde{T}_{ij}^{u^2}} \\ & \left. + P_7 \tilde{T}_{ij}^u \right], \quad (21) \end{aligned}$$

and P_d with $d = 1, \dots, 7$, is combinatorial constant corresponding to (1); P_7 is related to the magnitude of $\tilde{\mathbf{v}}_{ij}^{\perp u}$, i.e., if $\tilde{\mathbf{v}}_{ij}^{\perp u}$ is 0, P_7 is equal to $-\frac{W}{2} \sqrt{\frac{2W}{n_r \rho A}}$, otherwise P_7 is equal to 0. (17) and (18) correspond to the maximum UAV speed constraint as well as the maximum segment length constraint. $\Delta_{ij}^u \triangleq \tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u$ and $\Delta_{ij}^{hu} \triangleq \tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u$ define the projection lengths of the u th line segment in the horizontal and vertical directions, respectively.

According to the perspective operation preserver convexity [33], the third and seventh terms of (21) with respect to $\mathbf{Q}_1$, $\mathbf{H}_1$ and $\mathbf{T}_1$ are nonconvex, thus Problem (P2.1) is nonconvex.

B. SCA-Based Solution to Problem (P2.1)

To solve Problem (P2.1), slack variables are introduced to convert this nonconvex problem into a convex one, and the SCA technique is further applied to address it.

The convexity of the third and seventh terms in (21) is changed by introducing three slack variables, including $\mathbf{B} = \{B_{ij}^u \geq 0, \forall u \in \mathcal{U}\}$, $\mathbf{C} = \{C_{ij}^u \geq 0, \forall u \in \mathcal{U}\}$ and $\mathbf{D} = \{D_{ij}^u \geq 0, \forall u \in \mathcal{U}\}$, such that

$$B_{ij}^u = \sqrt{\sqrt{\tilde{T}_{ij}^{u^4} + \frac{\Delta_{ij}^{u^4}}{4v_0^4}} - \frac{\Delta_{ij}^{u^2}}{2v_0^2}}, \quad \forall u \in \mathcal{U}, \quad (22)$$

which is equivalent to

$$\frac{\tilde{T}_{ij}^{u^4}}{B_{ij}^{u^2}} = B_{ij}^{u^2} + \frac{\|\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u\|^2}{v_0^2}, \quad \forall u \in \mathcal{U}, \quad (23)$$

$$C_{ij}^u = \sqrt{\Delta_{ij}^{hu^2} + P_5 |\Delta_{ij}^{hu}| |\Delta_{ij}^{hu}| + P_6 \tilde{T}_{ij}^{u^2}}, \quad \forall u \in \mathcal{U}, \quad (24)$$

and

$$D_{ij}^u = C_{ij}^u \left(\frac{\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u}{\tilde{T}_{ij}^u} \right)^2, \quad \forall u \in \mathcal{U}. \quad (25)$$

Therefore, the third term of (21) can be replaced by the linear expressions $P_{in} B_{ij}^u$, with the additional Constraint (23). Besides, the seventh term of (21) can also replaced by $P_8 \left(\frac{C_{ij}^{u^3}}{\tilde{T}_{ij}^{u^2}} - D_{ij}^u \right)$, where P_8 is equal to $\frac{n_r A \rho}{4}$.

By substituting the above slack variables, Problem (P2.1) can be rewritten as

$$\begin{aligned} (P2.2): \min_{\mathbf{Q}_1, \mathbf{H}_1, \mathbf{T}_1, \mathbf{B}, \mathbf{C}, \mathbf{D}} & \sum_{u=0}^U \left[P_{bl} \tilde{T}_{ij}^u + P_1 \frac{\|\Delta_{ij}^u\|^2}{\tilde{T}_{ij}^u} + P_{in} B_{ij}^u \right. \\ & + P_2 \frac{\|\Delta_{ij}^u\|^3}{\tilde{T}_{ij}^{u^2}} + P_3 |\Delta_{ij}^{hu}| + P_4 \frac{\Delta_{ij}^{hu^3}}{\tilde{T}_{ij}^{u^2}} \\ & + P_8 \left(\frac{C_{ij}^{u^3}}{\tilde{T}_{ij}^{u^2}} - D_{ij}^u \right) + P_7 \tilde{T}_{ij}^u \left. \right], \\ \text{s.t.} \quad & \frac{\tilde{T}_{ij}^{u^4}}{B_{ij}^{u^2}} \leq B_{ij}^{u^2} + \frac{\|\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u\|^2}{v_0^2}, \\ & \forall u \in \mathcal{U}, \quad (26) \end{aligned}$$

$$C_{ij}^u \geq \sqrt{\Delta_{ij}^{hu^2} + P_5 |\Delta_{ij}^{hu}| \Delta_{ij}^{hu} + P_6 \tilde{T}_{ij}^{u^2}},$$

$$\forall u \in \mathcal{U}, \quad (27)$$

$$D_{ij}^u \leq C_{ij}^u \left(\frac{\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u}{\tilde{T}_{ij}^u} \right)^2,$$

$$\forall u \in \mathcal{U}, \quad (28)$$

$$B_{ij}^u \geq 0, \quad C_{ij}^u \geq 0, \quad D_{ij}^u \geq 0,$$

$$\forall u \in \mathcal{U}, \quad (17)-(20). \quad (29)$$

Note that in Problem (P2.2), constraints (26)–(28) are derived from (23)–(25) by replacing equalities with inequalities. This does not affect the equivalence between Problem (P2.1) and Problem (P2.2). The reason is that there always exists an optimal solution to Problem (P2.2) that makes all constraints satisfied with strict equality.

Problem (P2.2) is still nonconvex since constraints (26) and (28) are nonconvex. However, using the global lower/upper bounds of at a given local point to update these two nonconvex constraints can induce them to be convex. Due to the fact that the first-order Taylor expansion is a global lower bound of a convex function, the following inequality can be obtained for the right hand side (RHS) convex function of the nonconvex constraint (26) as

$$B_{ij}^u + \frac{\|\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u\|^2}{v_0^2} \geq B_{ij}^{u(r)^2} + 2B_{ij}^{u(r)} (B_{ij}^u - B_{ij}^{u(r)})$$

$$- \frac{\|\tilde{\mathbf{q}}_{ij}^{u+1(r)} - \tilde{\mathbf{q}}_{ij}^{u(r)}\|^2}{v_0^2}$$

$$+ \frac{2}{v_0^2} (\tilde{\mathbf{q}}_{ij}^{u+1(r)} - \tilde{\mathbf{q}}_{ij}^{u(r)})^T$$

$$\times (\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u)$$

$$\triangleq a_{ij}^{u(r)} (B_{ij}^u, \tilde{\mathbf{q}}_{ij}^u), \quad (30)$$

where $B_{ij}^{u(r)}$ and $\tilde{\mathbf{q}}_{ij}^{u(r)}$ are the current values of the corresponding variables at the r th iteration.

Furthermore, it can be found that the RHS of the nonconvex Constraint (28) is a joint nonconvex function with respect to $C_{ij}^u, \Delta_{ij}^{hu}$ and $\tilde{T}_{ij}^u$. Following a similar analysis in the work [34], the first-order Taylor expansion is a global upper bound of nonconvex functions, hence it can be obtained that

$$C_{ij}^u \left(\frac{\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u}{\tilde{T}_{ij}^u} \right)^2 \leq \left(\frac{\tilde{h}_{ij}^{u+1(r)} - \tilde{h}_{ij}^{u(r)}}{\tilde{T}_{ij}^{u(r)}} \right)^2 C_{ij}^u + 2C_{ij}^{u(r)}$$

$$\times \frac{\tilde{h}_{ij}^{u+1(r)} - \tilde{h}_{ij}^{u(r)}}{\tilde{T}_{ij}^{u(r)^2}} (\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u$$

$$- \tilde{h}_{ij}^{u+1(r)} + \tilde{h}_{ij}^{u(r)}) - 2C_{ij}^{u(r)}$$

$$\times \frac{(\tilde{h}_{ij}^{u+1(r)} - \tilde{h}_{ij}^{u(r)})^2}{\tilde{T}_{ij}^{u(r)^3}} (\tilde{T}_{ij}^u - \tilde{T}_{ij}^{u(r)})$$

$$\triangleq b_{ij}^{u(r)} (C_{ij}^u, \tilde{h}_{ij}^u, \tilde{T}_{ij}^u), \quad (31)$$

where $\tilde{h}_{ij}^{u(r)}$, $\tilde{T}_{ij}^{u(r)}$ and $C_{ij}^{u(r)}$ are also the current values of corresponding variables at the r th iteration.

By replacing the RHS of the nonconvex constraints (26) and (28) with their corresponding lower bound $a_{ij}^{u(r)} (B_{ij}^u, \tilde{\mathbf{q}}_{ij}^u)$ and upper bound $b_{ij}^{u(r)} (C_{ij}^u, \tilde{h}_{ij}^u, \tilde{T}_{ij}^u)$ at the r th iteration, respectively, it can be verified that Problem (P2.2) is convex. Therefore, standard convex optimization techniques or existing software toolbox (e.g., CVX) can be used to solve Problem (P2.2) efficiently. The SCA-based algorithm is summarized in Algorithm 1.

C. Convexity Analysis of Inequality Constraints

The Hessian matrix is applied to confirm the convexity of inequality constraints. Concretely, confirming the convexity of standardized inequality constraints involves verifying the semi-definiteness of their LHS functions' Hessian matrices [35]; they are convex if semi-definite, and nonconvex otherwise. Since the work [11], [12] has analyzed the convexity of constraints similar to constraints (26) and (29), only the convexity of constraints (27) and (28) shall be discussed at this point.

Proposition 2: Constraint (27) is convex.

Proof: The LHS function of the standardized Constraint (27), denoted as $f(C_{ij}^u, \Delta_{ij}^{hu}, \tilde{T}_{ij}^u)$, can be expressed as

$$f(C_{ij}^u, \Delta_{ij}^{hu}, \tilde{T}_{ij}^u) = \sqrt{\Delta_{ij}^{hu^2} + P_5 |\Delta_{ij}^{hu}| \Delta_{ij}^{hu} + P_6 \tilde{T}_{ij}^{u^2}} - C_{ij}^u, \quad (32)$$

where $-C_{ij}^u$ is convex [33], and the convexity corresponding to the first term of (32), denoted as $f_1(\Delta_{ij}^{hu}, \tilde{T}_{ij}^u)$, is validated by the Hessian matrix.

For ease of description, Δ_{ij}^{hu} and $\tilde{T}_{ij}^u$ are replaced by $\mathbf{x}$ and $\mathbf{y}$, respectively, and the Hessian matrix of $f_1(\mathbf{x}, \mathbf{y})$, denoted as Hes_1 , is as follows

$$Hes_1 = \begin{bmatrix} \nabla_{\mathbf{xx}}^2 f_1 & \nabla_{\mathbf{xy}}^2 f_1 \\ \nabla_{\mathbf{yx}}^2 f_1 & \nabla_{\mathbf{yy}}^2 f_1 \end{bmatrix}. \quad (33)$$

By calculating the second-order derivative, the closed-form expression of Hes_1 is given as

$$\nabla_{\mathbf{xx}}^2 f_1 = \frac{P_6(1 + \text{sgn}(\mathbf{x})P_5)\mathbf{y}^2}{((1 + \text{sgn}(\mathbf{x})P_5)\mathbf{x}^2 + P_6\mathbf{y}^2)^{\frac{3}{2}}},$$

$$\nabla_{\mathbf{yy}}^2 f_1 = \frac{P_6(1 + \text{sgn}(\mathbf{x})P_5)\mathbf{x}^2}{((1 + \text{sgn}(\mathbf{x})P_5)\mathbf{x}^2 + P_6\mathbf{y}^2)^{\frac{3}{2}}},$$

$$\nabla_{\mathbf{xy}}^2 f_1 = \nabla_{\mathbf{yx}}^2 f_1 = \frac{P_6(1 + \text{sgn}(\mathbf{x})P_5)\mathbf{xy}}{((1 + \text{sgn}(\mathbf{x})P_5)\mathbf{x}^2 + P_6\mathbf{y}^2)^{\frac{3}{2}}}, \quad (34)$$

where $\nabla_{\mathbf{xx}}^2 f_1$ is greater than 0 due to that $|\text{sgn}(\mathbf{x})P_5|$ is less than 1 [36].

On this basis, the size of Hes_1 , defined as $|Hes_1|$, satisfies

$$|Hes_1| = \nabla_{xx}^2 f_1 \nabla_{yy}^2 f_1 - \nabla_{xy}^2 f_1 \nabla_{yx}^2 f_1 = 0. \quad (35)$$

Thus, Hes_1 is semi-definite on account of $\nabla_{xx}^2 f_1$ and $|Hes_1|$ being non-negative, and the constraint (27) is convex. $\square$

Proposition 3: Constraint (28) is nonconvex.

Proof: Likewise, as D_{ij}^u is convex, demonstrating that the negative of Constraint (28)'s RHS function, represented by $f_2(C_{ij}^u, \Delta_{ij}^{hu}, \tilde{T}_{ij}^u)$, is nonconvex can confirm the non-convexity of (28). This entails proving that the Hessian matrix of $f_2(C_{ij}^u, \Delta_{ij}^{hu}, \tilde{T}_{ij}^u)$, denoted as Hes_2 , is non-semi-definite, which can be expressed as

$$Hes_2 = \begin{bmatrix} \nabla_{aa}^2 f_2 & \nabla_{ab}^2 f_2 & \nabla_{ac}^2 f_2 \\ \nabla_{ba}^2 f_2 & \nabla_{bb}^2 f_2 & \nabla_{bc}^2 f_2 \\ \nabla_{ca}^2 f_2 & \nabla_{cb}^2 f_2 & \nabla_{cc}^2 f_2 \end{bmatrix}, \quad (36)$$

where $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ stand for C_{ij}^u , Δ_{ij}^{hu} and $\tilde{T}_{ij}^h$, respectively.

By computing $\nabla_{aa}^2 f_2$, $\nabla_{ab}^2 f_2$, $\nabla_{ba}^2 f_2$ and $\nabla_{bb}^2 f_2$, the size of Hes_2 's second-order principal subform, denoted as $|Hes_{2,2}|$, can be expressed in closed form as

$$\begin{aligned} |Hes_{2,2}| &= \nabla_{aa}^2 f_2 \nabla_{bb}^2 f_2 - \nabla_{ab}^2 f_2 \nabla_{ba}^2 f_2 \\ &= 0 \times \left(-\frac{2\mathbf{a}}{\mathbf{c}^2} \right) - \left(-\frac{2\mathbf{b}}{\mathbf{c}^2} \right) \times \left(-\frac{2\mathbf{b}}{\mathbf{c}^2} \right) \\ &= -\frac{4\mathbf{b}^2}{\mathbf{c}^4}. \end{aligned} \quad (37)$$

Since $|Hes_{2,2}|$ is negative, $|Hes_2|$ is non-semi-definite, thus Constraint (28) is nonconvex. $\square$

D. Performance Analysis of Algorithm 1

To further demonstrate the feasibility of Algorithm 1, the performance of Algorithm 1 is analyzed in terms of complexity and convergence, respectively.

1) *Convergence Analysis:* Inspired by convergence analysis in [37], it can be concluded that Algorithm 1 provides a sequence of non-increasing values over the iteration, and its convergence is guaranteed. The proof is as follows, Problem (P2.1) is transformed as Problem (P2.2); the nonconvex Problem (P2.1) is approximated with convex surrogate Problem (P2.2) in each iteration and Problem (P2.2) is solved iteratively. According to the characteristics of the SCA technique, it follows that

$$E_1(\mathbf{Q}_1^u, \mathbf{H}_1^u, \mathbf{T}_1^u) \geq E_1(\mathbf{Q}_1^{u+1}, \mathbf{H}_1^{u+1}, \mathbf{T}_1^{u+1}). \quad (38)$$

Therefore, the objective value of Problem (P2.1) is non-increasing. Since the objective value of Problem (P2.1) is lower bounded by a finite value, Algorithm 1 is guaranteed to converge.

2) *Complexity Analysis:* According to the analysis in [38], since Problem (P2.2) is solved by the CVX solver, the overall computational complexity of Algorithm 1 can be easily given as $\mathcal{O}(U^{3.5} \log \frac{1}{\epsilon})$, where ϵ is a given threshold that determines the termination of Algorithm 1, i.e., the algorithm halts when the percentage decrease in the objective values of the solutions generated during its execution falls below this threshold.

Remark 1: Based on Algorithm 1, the energy consumption of UAVs accessing GN i and GN j sequentially in linear flight

Algorithm 1: SCA-Based Algorithm.

- 1: **Initialization:** obtain a feasible $\{\tilde{\mathbf{q}}_{ij}^{u(0)}\}$, $\{\tilde{T}_{ij}^{u(0)}\}$, $\{\tilde{h}_{ij}^{u(0)}\}$, $\{B_{ij}^{u(0)}\}$, $\{C_{ij}^{u(0)}\}$ and $\{D_{ij}^{u(0)}\}$ of Problem (P2.2). Set $r = 0$.
- 2: **repeat**
- 3: Replace the right hand side of (26) and (28) with $a_{ij}^{u(r)}(B_{ij}^u, \tilde{\mathbf{q}}_{ij}^u)$ and $b_{ij}^{u(r)}(C_{ij}^u, \tilde{h}_{ij}^u, \tilde{T}_{ij}^u)$, respectively.
- 4: Solve the convex Problem (P2.2), and denote the optimal solution as $\{\tilde{\mathbf{q}}_{ij}^u\}$, $\{\tilde{T}_{ij}^u\}$, $\{\tilde{h}_{ij}^u\}$, $\{B_{ij}^u\}$, $\{C_{ij}^u\}$ and $\{D_{ij}^u\}$.
- 5: Update the local point $\tilde{\mathbf{q}}_{ij}^{u(r+1)} = \tilde{\mathbf{q}}_{ij}^u$, $\tilde{T}_{ij}^{u(r+1)} = \tilde{T}_{ij}^u$, $\tilde{h}_{ij}^{u(r+1)} = \tilde{h}_{ij}^u$, $B_{ij}^{u(r+1)} = B_{ij}^u$, $C_{ij}^{u(r+1)} = C_{ij}^u$ and $D_{ij}^{u(r+1)} = D_{ij}^u$.
- 6: Update $r = r + 1$.
- 7: **until** the percentage decrease of the objective value of (P2.2) is below a given threshold ϵ or the number of iterations reaches N_ϵ .

(which is considered in Section VI) can also be minimized by adding a constraint, i.e., $\|\tilde{\mathbf{q}}_{ij}^{u+1} - \tilde{\mathbf{q}}_{ij}^u\| \cdot |H_j - H_i| = \|\mathbf{w}_j - \mathbf{w}_i\| \cdot |\tilde{h}_{ij}^{u+1} - \tilde{h}_{ij}^u|$, $\forall u \in \mathcal{U}$, to Problem (P2.2).

V. MPSO AND GPA FOR SOLVING PROBLEM (P3)

This section provides a novel framework to solve Problem (P3) to obtain the minimum number of UAVs and the minimum total UAV energy consumption.

A. Preliminaries

Prior to presenting the proposed framework, a two-stage scheme for minimizing the number of UAVs designed in [39] is introduced. It is specified as follows, first, obtaining an optimal tour, which consists of the path points passed by a UAV with sufficient energy to fulfill tasks with the minimum flight distance; second, decomposing the optimal tour into several short tours, and ensuring that the UAV energy consumption of each short tour does not exceed a fixed energy consumption upper bound. More importantly, calculating the minimum number of these short tours to approximate the minimum number of UAVs.

Inspired by this, a three-stage framework based on MPSO and GPA is developed to solve Problem (P3). Concretely, in the first stage, the MPSO algorithm is proposed to obtain an optimal sequence, namely O^* , which minimizes the energy consumption required for all GNs to be accessed by a UAV. In the second stage, the GPA algorithm is designed to decompose O^* into M_p subsequences, i.e., $O_1, O_2, \dots, O_{M_p}$, the minimum M_p is served as the minimum number of UAVs. On this basis, in the third stage, given that Problem (P3) is approximated as a MTSP (see details in Proposition 1), the MPSO algorithm is reused to solve Problem (P3) to derive the minimum total UAV energy consumption.

In the following, the two algorithms involved in the proposed framework are described.

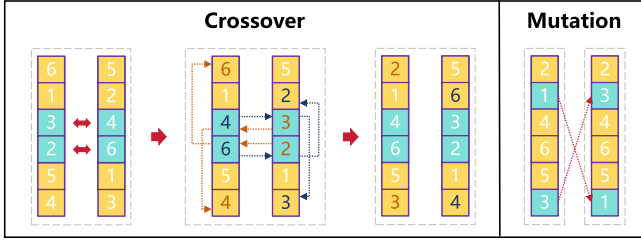


Fig. 3. Schematics of the crossover and mutation operations.

B. Modified Particle Swarm Optimization

The MPSO algorithm focuses on improving the search capability of the traditional PSO algorithm by introducing crossover and mutation operations from the genetic algorithm (GA). For ease of understanding, the MPSO algorithm is explained in four parts as follows. The overall algorithm is summarized in Algorithm 2.

1) *Initialization*: Initialization determines the initial population and the initial optimal solutions of the algorithm. The initial population, composed of L particles, denoted as P_a^l , $l = 1, \dots, L$, each with a GN-accessing sequence of dimension N , collectively forms the initial local optima (termed $lbest$); moreover, the global optima (termed $gbest$) are initialized as the particles with the smallest fitness value. The sequences of local optima and global optima, namely $lbest_{record}$ and $gbest_{record}$, respectively, are recorded.

2) *Fitness*: Fitness guides the optimization search direction of the algorithm. In the algorithm, particles' fitness is gauged by the energy required for UAVs to execute tasks corresponding to those particles' GN-accessing sequences, as calculated using (9). The particles with the smallest fitness value are chosen as the optimal particles in each iteration, concurrently, the GN-accessing sequences in these optimal particles are treated as the current optimal solutions.

3) *Crossover*: Crossover improves the global search capability of the algorithm. In each iteration, the partially matched crossover (PMX) strategy is employed to mix the current particles' solutions (i.e., the GN-accessing sequences) with $lbest_{record}$ (or $gbest_{record}$) to generate crossed particles. This approach preserves the superior traits of the parent particles while introducing new particles to increase diversity, broadening the global search range of the algorithm. See the left side of Fig. 3 for details.

4) *Mutation*: Mutation improves the local search capability of the algorithm. In each iteration, the GNs at any two positions in the crossed particles' GN-accessing sequences are swapped, introducing new variants. This prevents the algorithm from converging prematurely to local optima. On this basis, the particles with the smallest fitness value are chosen to update $gbest$, $lbest$, $gbest_{record}$ and $lbest_{record}$. See the right side of Fig. 3 for details.

Remark 2: The third stage's MPSO algorithm for solving MTSP, readily inferable from Algorithm 2, is omitted here for clarity.

Algorithm 2: MPSO Algorithm.

```

1: Input:  $L$  random particles  $P_a^l$ ,  $l = 1, \dots, L$ , the
   number of iterations  $it_n$ ,  $E_{\max}$ ,  $lbest$ ,  $lbest_{record}$ ,
    $gbest$ ,  $gbest_{record}$ .
2: Output: tour  $O^*$ .
3: while  $it_n \neq 0$  do
4:   for  $i \leftarrow 1$  to  $L$  do
5:     Cross  $P_a^i$  and  $lbest^i$ ;
6:     Judge constraints and calculate the fitness values
       by using (9);
7:     if  $E_{P_a^i} < E_{lbest^i}$  then
8:       Update  $E_{lbest^i}$  and  $lbest_{record}^i$ ;
9:     end if
10:    if  $E_{lbest^i} < E_{gbest^i}$  then
11:      Update  $E_{gbest^i}$  and  $gbest_{record}^i$ ;
12:    end if
13:    Cross  $P_a^i$  and  $gbest^i$ ;
14:    Repeat, calculate and update by using (9);
15:    Mutate  $P_a^i$ ;
16:    Repeat, calculate and update by using (9);
17:  end for
18:   $it_n \leftarrow it_n - 1$ ;
19: end while
20: Find minimum  $E_{gbest}^*$  and corresponding  $gbest_{record}^*$ ;
21:  $O^* \leftarrow gbest_{record}^*$ ;
22: return  $O^*$ ;

```

C. Greedy Path Assignment

In the GPA algorithm, the global optimal GN-accessing sequence obtained by the MPSO algorithm is divided into several local optimal GN-accessing subsequences via path assignment, and the number of local optimal subsequences represents the minimum number of UAVs. In each path assignment round, the GPA algorithm selects the first longest subsequence of the current global optimal sequence, with corresponding UAV energy consumption under $E_{\max}$, as the local optimal subsequence. Note that after each path assignment, the global optimal sequence is updated by removing the GNs in the local optimal subsequence selected in the current round.

The following lemma and theorem will further prove the feasibility of the GPA algorithm.

Lemma 1: The problem of minimizing the energy consumed by a UAV to access N GNs has the optimal substructure.

Proof: First, considering the optimal solution sequence, namely $O^* = \{g_1, g_2, \dots, g_N\}$, for this problem. It is assumed that for any $k \in [1, N]$, the set $G = \{g_1, g_2, \dots, g_k\}$, including the first k GNs, is proven to have an optimal substructure.

Then, incorporating g_{k+1} to the set G results in the expanded the set $G' = \{g_1, g_2, \dots, g_k, g_{k+1}\}$. Among all possible paths leading to g_{k+1} except for g_{k+1} itself, the path from g_k to g_{k+1} is the one with the minimum energy consumption. This is because the existence of any other GN g_i , $i \in \{1, \dots, k, k+2, \dots, N\}$ that offers a smaller energy consumption path to g_{k+1} would contradict the optimal O^* .

As a result, the set G' consisting of the first $k+1$ GNs is optimal and also has an optimal substructure. Therefore, this problem has an optimal substructure. $\square$

Theorem 1: The first longest subsequence satisfying the $E_{\max}$ constraint is the first optimal subsequence.

Proof: Following Lemma 1, the sequence consisting of any consecutive $k \in [1, N]$ GNs in the optimal GN sequence O^* is an optimal subsequence. Therefore, the first optimal subsequence containing k GNs is the first k terms of O^* . If k is related to $E_{\max}$ at this point, i.e., it satisfies conditions $E_{01} + \sum_{i=1, j=i+1}^{k-2} E_{ij} + E_{(k-1)0} \leq E_{\max}$, and $E_{01} + \sum_{i=1, j=i+1}^{k-1} E_{ij} + E_{k0} > E_{\max}$, the first longest subsequence satisfying the $E_{\max}$ constraint is the first k -term optimal subsequence. Note that i and j in E_{ij} correspond to the i th and j th terms in O^* at this point, respectively. $\square$

D. Performance Analysis of MPSO-GPA

In this subsection, the convergence analysis and the complexity analysis of MPSO-GPA are provided.

1) **Convergence Analysis:** The convergence of MPSO-GPA mainly depends on the convergence of the MPSO algorithm and the convergence of the GPA algorithm. Specifically, first, following the analysis of the traditional PSO algorithm in [40], it can be known that the convergence of the MPSO algorithm is mainly related to the particle updating, i.e., crossover and mutation; moreover, [41] has elaborated that crossover and mutation can guarantee the convergence of the genetic algorithm. Hence, the MPSO algorithm is convergent. Second, since the greedy-based algorithm is convergent [42], the GPA algorithm is convergent. Thus, MPSO-GPA is convergent.

2) **Complexity Analysis:** The computational complexity of MPSO-GPA is correspondingly associated with both the MPSO and GPA algorithms. In MPSO-GPA, the MPSO algorithm deals with a TSP and a MTSP, and the GPA algorithm deals with a path assignment problem. The two computational complexities corresponding to the former are $\mathcal{O}(N \cdot L \cdot it_n)$ and $\mathcal{O}(M \cdot N \cdot L \cdot it_n)$, respectively, while the latter corresponds to the computational complexity of $\mathcal{O}(N)$. Therefore, the computational complexity of MPSO-GPA is $\mathcal{O}(M \cdot N \cdot L \cdot it_n)$.

VI. SIMULATION SETUP AND RESULTS

In this section, simulations are performed to evaluate the effectiveness of the proposed methods. First, the simulation setup is given. Then, the minimum energy consumption and the minimum energy consumption rate of UAVs under linear trajectories and optimized trajectories are compared. Next, the performance of the MPSO algorithm and the MPSO-GPA algorithm in solving single-UAV and multi-UAV GN-accessing problems are evaluated, respectively, and finally, the feasibility of the overall algorithms is further emphasized. All methods are implemented in MATLAB and the detailed simulation results are presented as follows.

Algorithm 3: GPA Algorithm.

```

1: Input: tour  $O^*$ ,  $E_{\max}$ .
2:  $//O^*$  is obtained based on Algorithm 2 and expressed
   as  $\{g_0, g_1, g_2, \dots, g_N, g_0\}$ .
3: Output: minimum number of UAVs  $M_p$ .
4: Initialization:  $M_p \leftarrow 0$ ,  $i \leftarrow 1$  and  $j \leftarrow 1$ , where  $i$ 
   and  $j$  are indexes of the first and last GNs in the next
   tour, respectively.
5: while  $j \leq N$  do
6:    $O_{M_p} \leftarrow \{g_0, g_i, g_{i+1}, \dots, g_j, g_0\}$ ;
7:   if  $E_{O_{M_p}} \leq E_{\max}$  then
8:     if  $j == N$  then
9:        $//$  The last energy-efficient tour.
10:       $O_{M_p} \leftarrow \{g_0, g_i, g_{i+1}, \dots, g_N, g_0\}$ ;
11:    else
12:       $//$  Contain one more GN.
13:       $j \leftarrow j + 1$ ;
14:    end if
15:  else
16:     $j \leftarrow j - 1$ ;
17:     $O_{M_p} \leftarrow \{g_0, g_i, g_{i+1}, \dots, g_j, g_0\}$ ;
18:     $M_p \leftarrow M_p + 1$ ;
19:     $//$  Initialize the next energy-efficient tour.
20:     $O_{M_p} \leftarrow \{g_0, g_0\}$ ;
21:     $j \leftarrow j + 1$ ;
22:     $i \leftarrow j$ ;
23:  end if
24: end while
25: return  $M_p$ ;

```

A. Simulation Setup

The parameters involved in the simulation are illustrated as follows. The maximum flight altitude of the UAV $H_{\max}$ is set to 500 m, which is consistent with the altitude setting in [43]. Regarding GN-accessing, a large number of GNs are randomly deployed in a 1000 m $\times$ 1000 m square area, and the altitude required to access each GN is random. Regarding UAV trajectories, referring to the setup in [11], [12], $\Delta_{\max}^{\parallel}$ and $\Delta_{\max}^{\perp}$ are set to 30 m and 5 m, respectively, and the number of line segments U is set to 200. Regarding the MPSO algorithm, the maximum number of iterations is set to 1000, the initial population size is set to 200, and the crossover probability and mutation probability are 0.9 and 0.1, respectively. The parameter values related to the UAV propulsion energy consumption are derived from experimental measurements in our previous work [20] (see Table II). The maximum UAV horizontal speed $V_{\max}^{\parallel}$ and the maximum UAV vertical speed $V_{\max}^{\perp}$ are set to 30 m/s and 5 m/s [20], [44], respectively.

B. Performance Evaluation of the UAV Trajectory Optimization for Energy Minimization

The minimum energy consumption and the minimum energy consumption rate of a UAV consecutively accessing two GNs under the optimized trajectory and linear trajectory are analyzed.

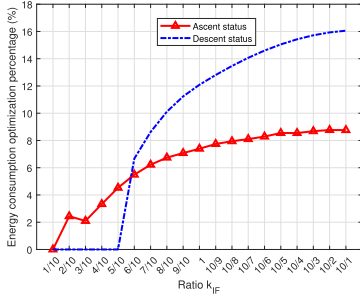


Fig. 4. Energy consumption optimization percentage versus k_{IF} for a UAV in two statuses.

First, two sets of GN pairs are designed to facilitate the discussion of the UAV trajectory in the ascent and descent statuses. Taking the ascent status as an example, the first term and the second term of a GN pair represent the initial and final points, respectively, and the corresponding 3-D coordinates of the first term are set to (0,0,0), while the magnitude of the 2-D coordinates and the height corresponding to the second term are set to $H_{\max}$ and $H_{\max}/k_{IF}$, respectively, where k_{IF} is defined as the ratio of the relative height and the relative horizontal distance between these two points. Moreover, the GN pairs designed for the UAV in the descent status are determined by interchanging the terms of the above GN pairs.

Then, to evaluate the difference of the minimum energy consumption for UAVs under these two trajectories, define $\frac{E_l^* - E_o^*}{E_l^*} \times 100\%$ to be the energy consumption optimization percentage, where E_l^* and E_o^* are the minimum energy consumption for UAVs under the linear trajectory and the optimized trajectory, respectively. Fig. 4 plots the energy consumption optimization percentages for the UAV in the ascent and descent flight statuses versus k_{IF} , respectively.

As can be seen, in most cases, the energy consumed by the UAV under the optimized trajectory is less than that consumed under the linear trajectory (i.e., the energy consumption optimization percentage is greater than 0). Specifically, in the ascent status, the energy optimization percentage is always greater than 0 and increases to around 8.5% with increasing k_{IF} before leveling off; in the descent status, the energy optimization percentage of the UAV is also not less than 0, and only when k_{IF} is less than 5/10, the energy optimization percentage of the UAV is equal to 0, indicating that the optimized trajectory of the UAV is linear at this time.

Next, in the same way, to evaluate the difference of the minimum energy consumption rates for UAVs under these two trajectories, define $\frac{E_l^*}{D_l} - \frac{E_o^*}{D_o}$ to be the energy consumption rate optimization, where D_l and D_o are the flight distances of UAVs under the linear trajectory and the optimized trajectory, respectively. Fig. 5 plots the energy consumption rate optimization for UAVs in the ascent and descent statuses versus k_{IF} , respectively.

It can be found that the energy consumption rate optimization of the UAV is always positively related to k_{IF} . Interestingly, given the same k_{IF} , the energy consumption rate optimization of the UAV in the ascent status is usually larger than that in the

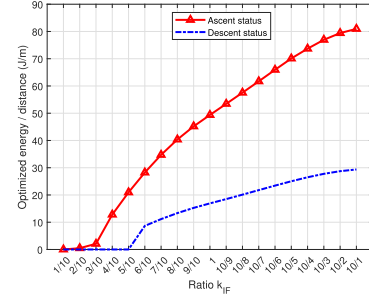


Fig. 5. Energy consumption rate optimization versus k_{IF} for a UAV in two statuses.

descent status since the UAV requires more energy in the ascent status [20]. This phenomenon becomes more significant as k_{IF} increases.

C. Performance Evaluation of MPSO-GPA

In order to fully evaluate the performance of MPSO-GPA, three *efficiencies* are analyzed from three perspectives. First, the efficiency of the MPSO algorithm and benchmarks is compared. Second, the efficiency of MPSO-GPA and the algorithms that integrate benchmarks and GPA is compared. Third, the efficiency of MPSO-GPA in two multi-UAV flight planning problems is compared. The following six optimization algorithms are introduced as benchmarks:

- Genetic Algorithm (GA) [45]: GA is an efficient and parallel global search algorithm.
- Branch-and-Bound (BnB) [17]: BnB is a popular exact algorithm that searches for the optimal solution by constructing a search tree.
- Nearest Neighbour (NN) [18]: NN is the simplest heuristic algorithm and usually has a fast execution speed.
- Particle Swarm Optimization (PSO) [46]: PSO is a simple algorithm and has a fast search speed.
- Modified Ant Colony Optimization (MACO) [47]: MACO is a varied type of the ACO algorithm characterized by superior search efficacy and faster convergence speed.
- Modified Genetic Algorithm (MGA) [48]: MGA is a varied type of GA characterized by a stronger global search capability.

First, considering energy-efficient single-UAV path planning problems with different GN sizes, the performance of the MPSO algorithm and benchmarks in solving these problems is evaluated. The size of GNs varies from 30 to 300, and the position of each GN is random, the maximum energy each UAV can utilize is set to $E_{\max} = \infty$. To improve the reliability of the simulation results, each simulation is repeated 30 times, and the average value is shown in Fig. 6.

As illustrated in Fig. 6(a), the MPSO algorithm minimizes energy consumption better than benchmarks. Specifically, compared with the PSO algorithm, the energy optimization efficiency of the MPSO algorithm is improved by an average of around 46.07%. Interestingly, it can be observed from Fig. 6(b) that the algorithms with better energy consumption optimization also

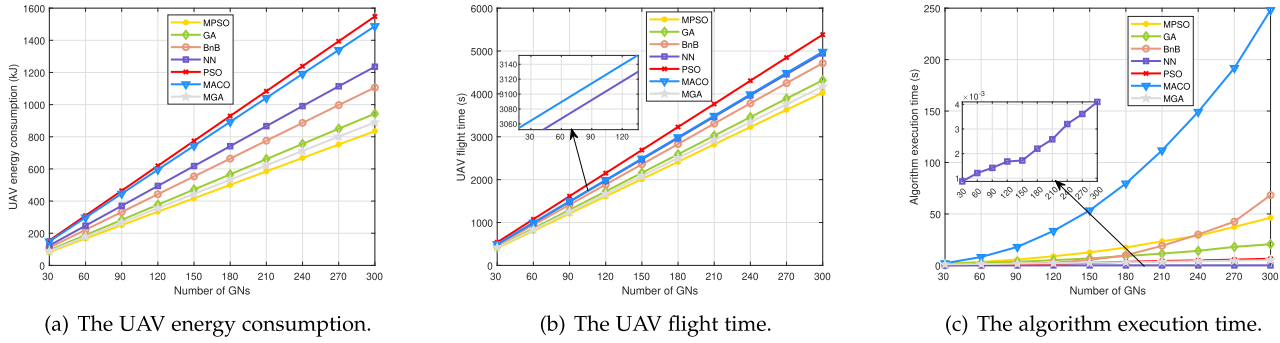


Fig. 6. Algorithm performance comparison for single-UAV path planning problems.

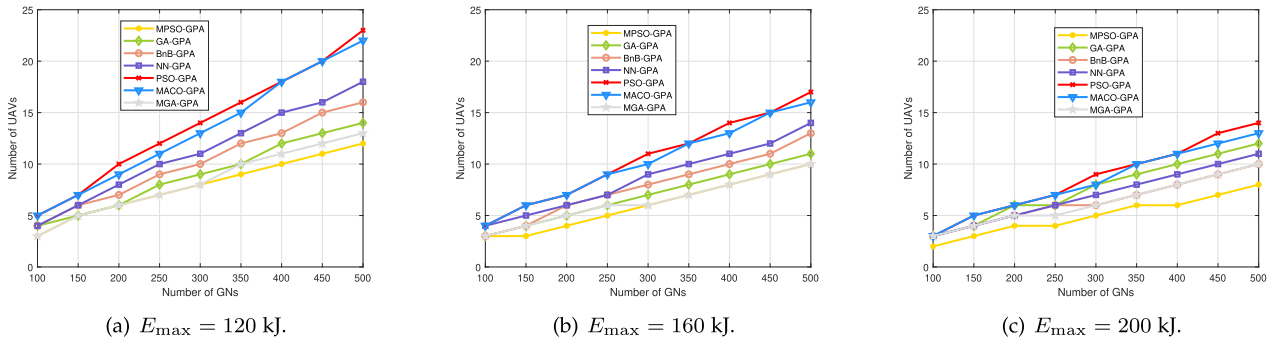


Fig. 7. Algorithm performance comparison for the minimum number of UAVs in multi-UAV path planning problems with different $E_{\max}$.

tend to derive less UAV flight time; meanwhile, the energy consumption and the flight time of UAVs obtained by these algorithms show an approximately linear relationship proportional to the number of GNs, reflecting the stability of these algorithms. In addition, it can be intuitively found from Fig. 6(c) that the MACO algorithm consistently exhibits the longest execution time, while the simpler NN algorithm maintains the shortest duration. Apart from the aforementioned extremes, the MPSSO algorithm consistently ranks high in execution time, only outpaced by the BnB algorithm when the number of GNs exceeds 240. Although the execution time of the MPSSO algorithm is not small, this high computational complexity in exchange for the significant reduction in energy consumption and flight time of UAVs makes the MPSSO algorithm more feasible in practice.

Second, considering energy-efficient multi-UAV path planning problems with different GN sizes, the results obtained by the seven integrated approximation algorithms in solving these problems are presented. Following the similar setup as above (modifications are restricted to the GN range from 100 to 500), the minimum number of UAVs in three $E_{\max}$ cases, i.e., 120 kJ, 160 kJ, and 200 kJ, are shown in Fig. 7, and the bars in Fig. 8 record corresponding energy consumption and flight time of UAVs. Expectedly, in terms of energy consumption and flight time, the stability of algorithms such as the MPSSO algorithm and GA pushes the frameworks based on them to maintain similar performance in multi-UAV path planning; only GA-GPA shows fatigue with increasing GN size. Therefore, more attention is paid to analyzing the minimum number of UAVs.

In Fig. 7(a), the good performance of the MPSSO algorithm leads MPSSO-GPA to minimize the number of UAVs efficiently as expected. As $E_{\max} = 120$ kJ, this phenomenon becomes more significant as the number of GNs increases.

Interestingly, it can be found from Fig. 7(b) and (c) that the minimum number of UAVs required remains constant when the number of GNs is within some specific interval. For example, when $E_{\max} = 160$ kJ and the number of GNs is between 100 and 150, the minimum number of UAVs obtained by MPSSO-GPA remains constant; when $E_{\max} = 200$ kJ and the number of GNs is between 200 and 250, the minimum number of UAVs obtained by GA-GPA remains constant. The reason is that the increase of $E_{\max}$ expands the coverage of UAVs.

Overall, regarding the three $E_{\max}$ cases, the efficiency of MPSSO-GPA in minimizing the number of UAVs compared to other algorithms is up to around 47.83%, 50% and 46.15%, respectively.

Lastly, to further highlight the performance of MPSSO-GPA, the minimum energy consumption and the minimum number of UAVs obtained by MPSSO-GPA in solving two multi-UAV path planning problems, i.e., considering energy consumption costs and considering path distance costs, are compared. The results (see Table III) show that the former involves more than 50% reduction in both the energy consumption and the number of UAVs after optimization compared to the latter. In addition, the optimization efficiency of MPSSO-GPA is less affected by the fluctuation of the GN number, verifying the robustness of the algorithm.

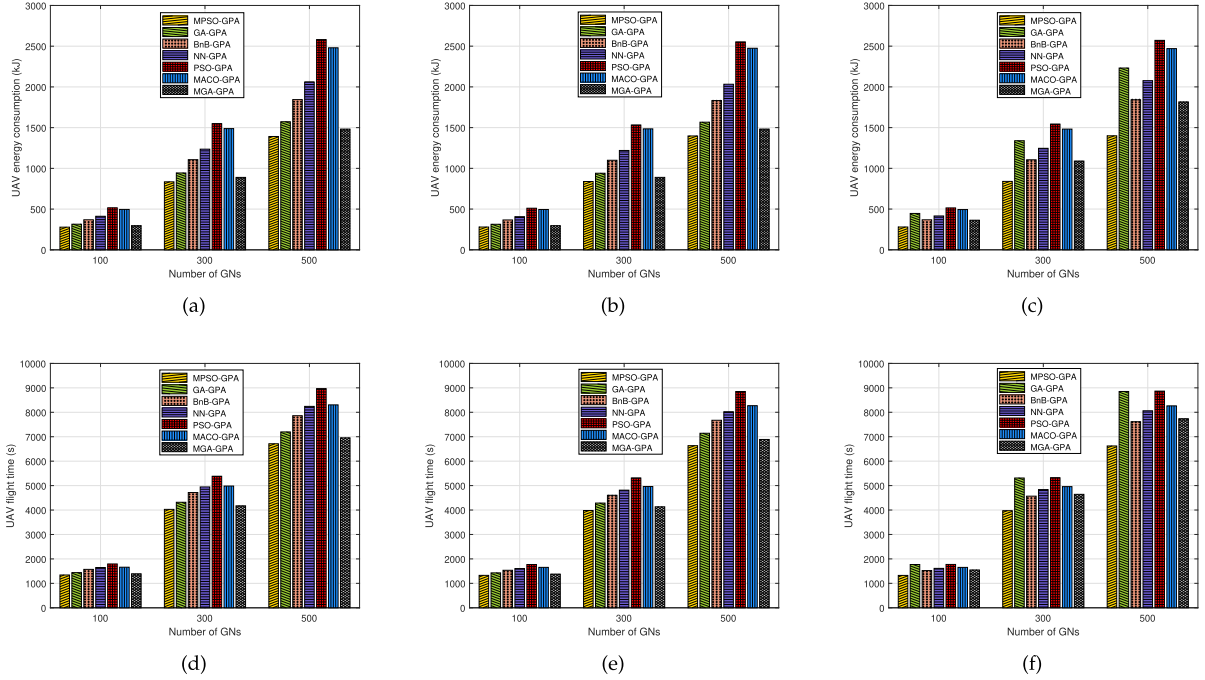


Fig. 8. Mean UAV energy consumption and mean UAV flight time for three $E_{\max}$ cases in multi-UAV flight planning problems ((a), (b) and (c) represent the mean energy consumption for three $E_{\max}$ cases, i.e., 120 kJ, 160 kJ, and 200 kJ, respectively, likewise, (d), (e), and (f) correspond to the mean flight time for each of these three cases).

TABLE III
MEAN UAV ENERGY CONSUMPTION AND MEAN NUMBER OF UAVS REQUIRED FOR TWO MULTI-UAV PATH PLANNING PROBLEMS

Instance		Energy (kJ)		Number	
num	N	$E_{\max}$ (kJ)	energy-based	distance-based	energy-based
1	100	120	279.3	670.7	3
2	100	160	278.2	671.8	2
3	100	200	279.8	675.3	2
4	200	120	558.6	1341.4	6
5	200	160	556.3	1343.6	4
6	200	200	559.7	1350.7	4
7	300	120	837.9	2012.1	8
8	300	160	834.5	2015.5	6
9	300	200	839.5	2026	5
10	400	120	1117.1	2680.8	10
11	400	160	1111.5	2685.9	8
12	400	200	1120.3	2702.9	6
13	500	120	1390.8	3353.5	12
14	500	160	1390.8	3359.1	10
15	500	200	1399.1	3376.7	8

D. Performance Evaluation of the Overall Algorithms

Synthesizing the above analysis, an example is used to corroborate the overall algorithms' utility in solving energy-efficient GN-accessing tasks, with subsequent analysis of UAV energy consumption and UAV altitude therein. First, the process of such a task is outlined in Fig. 9. To be specific, this GN-accessing task involves 15 GNs, the 3-D coordinates of the depot and $E_{\max}$ are set to (550,475,250) and 90 kJ, respectively. It is calculated that at least two UAVs, namely UAV 1 and UAV 2, are required to fulfill this task, where UAV 1 covers the GNs numbered 1 to 7 along the red curve, and UAV 2 covers the GNs numbered 8 to

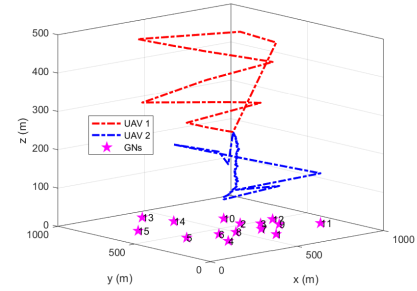


Fig. 9. 3-D trajectories of UAVs.

15 along the blue curve. The details of the two UAVs' flying paths are given in Table IV.

Then, the energy consumption and the flight altitude of UAVs 1 and 2 on their respective trajectories (i.e., the red and blue curves in Fig. 9) are presented.

Energy-efficient multi-UAV GN-accessing manifests in the reasonable utilization of UAV energy. As can be seen in Figs. 10 and 11, despite their distinct tasks, both UAV 1 and UAV 2 adopt a moderate onboard storage utilization approach, which is characterized by controlling UAV energy consumption to maintain small fluctuations, as evidenced during their respective trajectories from segment 0 to 1400 and segment 200 to 1800. This approach not only mitigates hardware wear in UAVs caused by frequent energy consumption variations, but also reduces extra energy consumption by decreasing the number of vertical flight status alterations, thereby enhancing GN access efficiency.

The flight altitude has an impact on the UAV energy consumption. Figs. 10 and 11 reveal a trend where larger altitude

TABLE IV
FLYING PATHS OF THE TWO UAVS

UAV No.	Number of GNs	Flying path
1 (red)	7	(550, 475, 250)→(260, 450, 300)→(700, 465, 315)→(400, 900, 300)→(800, 500, 410)→(200, 700, 500)→(635, 525, 500)→(600, 250, 500)→(550, 475, 250)
2 (blue)	8	(550, 475, 250)→(650, 650, 50)→(750, 400, 100)→(500, 400, 110)→(900, 300, 130)→(625, 375, 175)→(450, 750, 200)→(415, 415, 215)→(350, 280, 200)→(550, 475, 250)

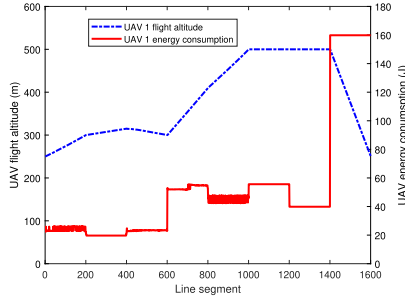


Fig. 10. UAV 1's instantaneous altitude and energy consumption versus line segments.

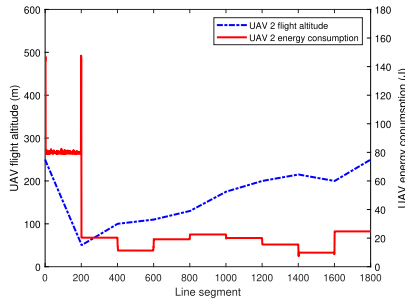


Fig. 11. UAV 2's instantaneous altitude and energy consumption versus line segments.

variations in UAV flights are associated with higher energy consumption. This is understandable since UAV vertical energy consumption is a major component of total UAV flight energy consumption, significant altitude changes leading to increased vertical energy consumption are directly reflected in increased total energy consumption [20]. Interestingly, in some special cases, the change in UAV flight altitude can cause transient changes in UAV energy consumption, e.g., the rapid decrease in energy consumption of UAV 2 at segment 200. This is mainly because the change in UAV flight status at this time brings a large energy consumption difference. However, these energy consumption transients are negligible in the long-term steady flight of UAVs.

VII. CONCLUSION

This paper proposed a generic multi-UAV application scenario, namely the GN-accessing scenario, and formulated a problem to minimize the energy consumption for GN-accessing tasks while employing the minimum number of UAVs. To solve this problem, this paper decomposed it into two subproblems. The first subproblem is to minimize the energy required for UAVs to access any two GNs consecutively, which was solved

by applying the SCA technique and path discretization method. The second subproblem is to plan optimal energy-efficient GN-accessing paths employing the minimum number of UAVs, which was solved by designing a three-stage approximation framework based on MPSTO and GPA. The simulation results showed that the UAV flying with the optimized trajectory can effectively reduce energy consumption by up to about 16% compared to the linear trajectory; in addition, the proposed MPSTO-GPA algorithm has more than 45% improvement in minimizing the number of UAVs compared to benchmarks. In the future, improving the MPSTO algorithm and updating the energy consumption model for UAV acceleration/deceleration will be further considered.

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